

1.

Which of the following operations could cause an error if the stack is initially empty?

- A. pop()
- B. push(item)
- C. isEmpty()
- D. size()

Answer: A

2.

If a stack has a capacity of 5 and elements 10, 2, 53, 94 are pushed onto it, what happens if you try to push 65 and then 56?

- A. Both pushes succeed.
- B. Pushing 65 succeeds, pushing 56 causes stack overflow.
- C. Pushing 56 succeeds, pushing 65 causes stack overflow.
- D. Pushing 65 causes stack overflow.

Answer: B

3.

A stack is initially empty. The following operations are performed: PUSH(5), PUSH(55), PUSH(5), POP(), PUSH(55), POP(), POP(), PUSH(51), POP(), What is at the top of the stack?

- | | |
|-------|-------|
| A. 55 | C. 51 |
| B. 15 | D. 5 |

Answer: D

4.

What is the time complexity of Pop operations in a stack implemented using an array of n elements?

- A. $O(n)$
- B. $O(1)$
- C. $O(2n)$
- D. $O(\log n)$

Answer: B

5.

The 'undo' and 'redo' functions in a text editor are commonly implemented using which data structure?

- A. Stack
- B. Queue
- C. Priority Queue
- D. Heap

Answer: A

6.

The prefix form of the infix expression $X * (Y - Z) / V - W$ is:

- A. $/ - * X - YZVW$
- B. $- / * X - YZVW$
- C. $/ - * X - YZWV$
- D. $/ - * Y - ZXWV$

Answer: B